

The d- and f-Block Elements

NCERT Class XII Chemistry · Chapter 4 · d and f Block Elements

d and f Block — Complete Notes

*11 marks, and material you haven't met before. Everything here is written from scratch — terms tagged [exposure] are ones you're seeing for the first time, and each gets a proper explanation before it's used. Almost every answer in this chapter traces back to one idea: how many unpaired d electrons are there?**

What the d block is — where these elements sit, and why they're called "transition"

[exposure] d-block elements are the elements of groups 3 to 12 — the wide middle block of the periodic table, sitting between the s block on the left and the p block on the right. What defines them is that electrons are filling a **d orbital**. Each atom has an outermost shell; the shell just inside it is the **penultimate** one, and it's that shell's d orbital being filled. General configuration:

$$(n-1)d^{1-10} ns^{1-2}$$

So if the outermost shell is the 4th, the 3d orbital is filling. There are four such series — 3d, 4d, 5d, 6d.

[exposure] Transition element is a narrower term with a precise definition: an element with a **partially filled d orbital**, either in its atom *or* in any of its stable oxidation states. The word "transition" comes from their position, bridging the reactive metals of the s block and the non-metals of the p block. The definition matters because it excludes some d-block members — see zinc below. Transition elements share a family of properties: variable oxidation states, coloured ions, catalytic behaviour, complex formation, and hardness.

The four series

Series	Orbital	Runs from	Period
First	3d	Sc (21) → Zn (30)	4th
Second	4d	Y (39) → Cd (48)	5th
Third	5d	La (57), then Hf (72) → Hg (80)	6th
Fourth	6d	Ac (89), then Rf (104) → Cn (112)	7th

The gaps in the third and fourth series (58–71 and 90–103) are where the f-block elements sit — pulled out and studied separately.

3d series configurations — learn these, most questions start here

All are $[\text{Ar}] 3d^n 4s^2$ except two exceptions:

Sc $3d^1 4s^2$	Ti $3d^2 4s^2$	V $3d^3 4s^2$
Cr $3d^5 4s^1$	Mn $3d^5 4s^2$	Fe $3d^6 4s^2$
Co $3d^7 4s^2$	Ni $3d^8 4s^2$	Cu $3d^{10} 4s^1$
Zn $3d^{10} 4s^2$		

Why Cr and Cu break the pattern: a **half-filled** (d^5) or **completely filled** (d^{10}) d subshell is extra stable, so an electron shifts from 4s to 3d to reach it.

Why Zn, Cd, Hg are not transition elements

Their configuration is $(n-1)d^{10}ns^2$ — the d orbital is **completely filled**, in the ground state and in their common +2 oxidation state (Zn^{2+} is $3d^{10}$). No partially filled d orbital in any stable state means they fail the definition.

Examiner asks: the flip side — "why is Cu a transition element despite a filled d orbital in the ground state?" Because its common oxidation state Cu^{2+} is $3d^9$, which *is* partially filled. Same for Sc: $3d^1$ in the ground state, so it qualifies even though Sc^{3+} is colourless.

Enthalpy of atomization — *the unpaired-electron idea, introduced properly*

[exposure] Enthalpy of atomization is the energy needed to take one mole of a solid metal and pull it apart into free gaseous atoms. It exists as a measure because it tells you directly how strong the metallic bonding is — strong bonds mean the atoms resist being separated, so the value is high. Transition metals have unusually high values: unpaired d electrons participate in metallic bonding, and more unpaired electrons means a stronger bond. Chromium, with six unpaired electrons ($3d^5 4s^1$), sits near the top; zinc, with none ($3d^{10} 4s^2$), is the lowest in the 3d series.

Two trends carry nearly all the marks:

- **Maximum in the middle of each series**, where unpaired electrons peak.
- **Second and third series > first series.** The 4d and 5d elements have poorly shielding inner electrons, so the effective nuclear charge is higher, metal-metal bonding is stronger, and atomization needs more energy.

Examiner asks: "zinc has the lowest enthalpy of atomization" (fully filled d, no unpaired electrons, weak metallic bond) and "why do 2nd/3rd series have greater enthalpy of atomization" (poor shielding → stronger metallic bond). Both recur.

Melting and boiling points follow the same logic

Highest in the middle: Cr in the 3d series, Mo in 4d, W in 5d. Mn and Tc are anomalously **low** — their $d^5 s^2$ configuration is so stable that the electrons are less free to delocalise into metallic bonding.

Zn, Cd, Hg are **soft metals with low melting points** for the same reason as their low atomization enthalpy: all d electrons paired, weak metallic bond.

Ionization enthalpy and the exchange-energy idea — *why the trend is so irregular*

Ionization enthalpy depends on three competing things:

1. **Nuclear attraction** — higher nuclear charge holds electrons tighter, raising it.
2. **Electron repulsion (screening)** — electrons pushing each other out lowers it.
3. **Exchange energy** — see below.

[exposure] Exchange energy is a stabilising effect that appears when several electrons with **parallel spins** occupy orbitals of equal energy. Such electrons can swap positions with each other, and every possible swap releases a little energy. More possible swaps means more energy released, and losing energy means greater stability. This is the real reason d^5 and d^{10} configurations are so stable — they maximise the number of parallel-spin exchanges. It's why chromium is $3d^5 4s^1$ rather than $3d^4 4s^2$.

The comparisons they set

- **Mn⁺ has lower IE than Cr⁺.** Mn⁺ is $3d^5 4s^1$; losing one electron gives the stable $3d^5$ Mn²⁺, so it goes easily. Cr⁺ is already $3d^5$; removing an electron breaks that stability.
- **Fe²⁺ has lower IE than Mn²⁺.** Fe²⁺ ($3d^6$) loses one to reach the stable $3d^5$; Mn²⁺ is already $3d^5$ and resists.
- **Second IE of Cr and Cu is high.** Cr⁺ ($3d^5$) and Cu⁺ ($3d^{10}$) are both stable, so the next electron is hard to remove.
- **Third IE of Mn and Zn is high** — Mn²⁺ ($3d^5$) and Zn²⁺ ($3d^{10}$) are stable.

Trap: every one of these is answered by writing the configuration *before and after* the electron leaves, and pointing at whichever side is d^5 or d^{10} . Don't reason in words alone — write both configurations.

Variable oxidation states — *the defining property, and why P-block differs*

[exposure] Variable oxidation state means one element can lose different numbers of electrons in different compounds — iron exists as both Fe²⁺ and Fe³⁺, manganese runs from +2 all the way to +7. It happens because in transition elements the $(n-1)d$ and ns orbitals are **very close in energy**. With so small a gap, electrons from both can be pulled into bonding, and the number that participates can vary. Main-group elements have no such near-degenerate pair, so their oxidation states are far more restricted.

- **Most common state is +2** across the 3d series (from losing the two 4s electrons) — except Sc, which is +3.
- **Maximum variety in the middle:** Mn shows +2 to +7, the most of any 3d element, because it has the most unpaired electrons available.
- **Fewest at the ends:** Sc and Ti have too few electrons to lose; Cu and Zn have too few vacant d orbitals to use.
- **Cu shows +1** ($3d^{10}$, stable) — the only 3d element with a common +1 state.

Transition vs p-block variability — a standing question

Transition metal oxidation states differ by **one** unit (Fe^{2+} , Fe^{3+} ; V shows 2, 3, 4, 5) because both $(n-1)d$ and ns contribute. P-block states differ by **two** (Sn^{2+} , Sn^{4+}) because the p orbital dominates and electrons are lost in pairs.

Zero oxidation state in carbonyls

In $\text{Ni}(\text{CO})_4$ and $\text{Fe}(\text{CO})_5$ the metal is in oxidation state **zero**. CO is neutral and forms a sigma bond to the metal while accepting d-electron density back as a pi bond — so no charge is transferred overall.

Standard electrode potentials — *reading the E° table for this series*

Across the 3d series, $E^\circ(\text{M}^{2+}/\text{M})$ becomes **less negative** — the tendency to form the +2 ion decreases — because ionization enthalpies rise as nuclear charge increases across the row.

Copper is the exception — positive E°

$E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$, meaning copper prefers to stay as the solid metal. Converting $\text{Cu}(\text{s})$ to Cu^{2+} needs a high enthalpy of atomization *plus* a high sum of first and second ionization enthalpies, and the energy released on hydration doesn't compensate.

Consequence asked directly: copper does not displace hydrogen from acids. It lies below hydrogen in the electrochemical series with a positive reduction potential, so it can't reduce H^+ to H_2 . Only oxidising acids (HNO_3 , hot conc. H_2SO_4) dissolve it.

Three more anomalies, each with the same explanation shape

- **Mn, Ni, Zn are more negative than the trend predicts.** Mn^{2+} is $3d^5$ (half-filled, stable); Zn^{2+} is $3d^{10}$ (filled, stable); Ni^{2+} is small so its hydration enthalpy is exceptionally large and negative.
- $E^\circ(\text{Mn}^{3+}/\text{Mn}^{2+})$ is **highly positive** — Mn^{3+} ($3d^4$) readily gains an electron to become the stable $3d^5$. So Mn^{3+} is a **good oxidising agent**.
- $E^\circ(\text{Cr}^{3+}/\text{Cr}^{2+})$ is **negative** — Cr^{2+} ($3d^4$) readily loses an electron to become $3d^3$, which is stable (t_{2g}^3 , half-filled in the lower set). So Cr^{2+} is a **strong reducing agent**.

Trap: Mn^{3+} oxidising and Cr^{2+} reducing are a matched pair, both answered by "which side is the stable configuration". Learn them together — the exam often asks for both in one question.

Cu^+ is **unstable in aqueous solution** — it disproportionates to Cu^{2+} and Cu , because the much more negative hydration enthalpy of the smaller Cu^{2+} more than compensates for the second ionization enthalpy.

Magnetic properties — *and the one formula you'll actually compute*

[exposure] Paramagnetic substances contain **unpaired electrons** and are weakly *attracted* into a magnetic field. **Diamagnetic** substances have all electrons paired and are weakly

repelled. The distinction exists because an unpaired electron behaves like a tiny magnet whose field isn't cancelled by a partner. Most transition metal ions have unpaired d electrons, so most of their compounds are paramagnetic — Mn^{2+} ($3d^5$, five unpaired) is strongly so, while Zn^{2+} ($3d^{10}$) is diamagnetic.

The strength is measured as **spin-only magnetic moment**:

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

n = number of unpaired electrons; BM = Bohr magneton.

Ion	Config	n	μ (BM)
Sc^{3+}	$3d^0$	0	0
Ti^{3+}	$3d^1$	1	1.73
V^{3+}	$3d^2$	2	2.83
Cr^{3+}	$3d^3$	3	3.87
Fe^{2+}	$3d^6$	4	4.90
Mn^{2+}	$3d^5$	5	5.92

Worked · 2020 — spin-only moment of Co^{2+}

Co has $Z = 27$: $[\text{Ar}] 3d^7 4s^2$. Removing two electrons (the 4s pair) gives $\text{Co}^{2+} = [\text{Ar}] 3d^7$.

Filling five d orbitals with 7 electrons: two orbitals paired, three singly occupied $\rightarrow n = 3$.

$$\mu = \sqrt{3(3+2)} = \sqrt{15} = 3.87 \text{ BM}$$

Trap: the 4s electrons come off **first**, before any 3d electron. Co^{2+} is $3d^7$, not $3d^5 4s^2$.

Why transition compounds are coloured — *d-d transition, and the one exception to it*

[exposure] A **d-d transition** is what happens when an electron absorbs light and jumps from a lower-energy d orbital to a higher-energy one. In a complex, the five d orbitals no longer all have the same energy — they split into two sets (called t_{2g} and e_g , covered properly in Coordination Compounds). The gap between them happens to match visible light, so the compound absorbs one colour and we see the **complementary** colour. Cu^{2+} ($3d^9$) has an unpaired electron available to jump, so its salts are blue; Zn^{2+} ($3d^{10}$) has no vacancy to jump into, so its salts are white.

The rule: colour requires **partially filled** d orbitals. Both d^0 and d^{10} are colourless.

Ion	Config	Colour
Sc ³⁺ , Ti ⁴⁺	3d ⁰	Colourless
Ti ³⁺	3d ¹	Purple
Cr ³⁺	3d ³	Green
Mn ²⁺	3d ⁵	Pink
Fe ²⁺	3d ⁶	Green
Cu ²⁺	3d ⁹	Blue
Zn ²⁺	3d ¹⁰	Colourless

The exception: charge transfer

[exposure] Ligand-to-metal charge transfer is a second, different way a compound can be coloured — an electron jumps from the surrounding ligand onto the metal, rather than between two d orbitals. It matters because it explains the compounds that are intensely coloured despite having *no* d electrons at all. KMnO₄ is the standard case: Mn is +7, so 3d⁰, and no d-d transition is possible — yet it's deep purple, because charge transfers from oxygen's 2p orbital to manganese's 3d. The same explains K₂Cr₂O₇'s orange.

Examiner asks: "why is KMnO₄ coloured despite 3d⁰?" — charge transfer, not d-d. Getting this distinction right is a whole mark.

Catalysis, interstitial compounds, alloys — *three properties, three clean explanations*

Catalytic behaviour

Two reasons, and good answers give both:

1. **Variable oxidation state** lets the metal form intermediates by changing state, then revert. The classic: Fe³⁺ catalysing the iodide-persulphate reaction — Fe³⁺ oxidises I⁻ to I₂ and becomes Fe²⁺, which is then re-oxidised by persulphate back to Fe³⁺.
2. **Surface adsorption** — reactants stick to the metal surface, which raises their local concentration and weakens their bonds, making reaction easier.

V₂O₅ is asked specifically: large surface area, and it changes oxidation state easily to form an unstable intermediate, providing a lower-energy path.

Interstitial compounds

[exposure] An **interstitial compound** forms when small atoms — hydrogen, boron, carbon, nitrogen — get trapped in the gaps between metal atoms in a crystal lattice. Transition metals sit in three-dimensional lattices with spaces (**interstitial sites**) between the atoms, and small atoms slot into them without displacing anything. The result isn't a normal compound: the ratio isn't a whole number, so these are called **non-stoichiometric**. VH_{0.56} means 56 hydrogen atoms per 100 vanadium atoms; TiH_{1.7} means 170 per 100.

Properties: higher melting points than the pure metal, very hard (some borides approach diamond), retain metallic conductivity, and are chemically inert.

Alloy formation

[exposure] An **alloy** is a homogeneous solid solution of two or more metals — melt them together and let them solidify. It forms readily when the metals' atomic radii are within about **15%** of each other, because then one metal's atoms can simply take the other's positions in the crystal lattice. Transition metals have very similar atomic sizes, so they alloy easily. Ferrous alloys (Fe with Cr, V, W, Mo, Mn) give steel and stainless steel; brass is Cu-Zn and bronze is Cu-Sn, both pairing a transition with a non-transition metal.

Examiner asks: "why do transition metals form alloys" — similar atomic size, so one metal replaces the other in the lattice. Two lines, two marks.

Oxides and oxoanions — *the acid-base trend that gets asked every year*

The **highest oxidation number in an oxide coincides with the group number** from Sc to Mn: Sc_2O_3 (+3, group 3) up to Mn_2O_7 (+7, group 7). Beyond group 7 no higher oxide than Fe_2O_3 is known.

The acid-base rule

Oxidation state of metal	Oxide is	Chromium example
Low	Basic	CrO (+2)
Intermediate	Amphoteric	Cr_2O_3 (+3)
High	Acidic	CrO_3 (+6)

Why: in a low oxidation state the metal has d electrons available to *donate*, so it behaves as a Lewis base. In a high oxidation state the large positive charge lets it *accept* an electron pair — a Lewis acid. As oxidation state rises, metallic character falls and covalent character rises.

Examiner asks: " Mn_2O_3 is basic whereas Mn_2O_7 is acidic" (2019) — a direct application. Give the oxidation numbers (+3 and +7) and the Lewis reasoning.

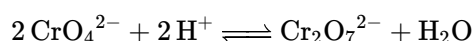
Oxygen beats fluorine at stabilising high oxidation states

Manganese's highest fluoride is MnF_4 (+4), but its highest oxide is Mn_2O_7 (+7) — even though fluorine is more electronegative. The reason: **oxygen can form multiple bonds** with the metal, fluorine only single bonds. More bonds per atom means a higher oxidation state can be supported.

Potassium dichromate — *preparation, the pH equilibrium, structures, oxidising action*

The pH interconversion — the most-asked single fact

Chromate and dichromate exist in equilibrium, and pH decides which:



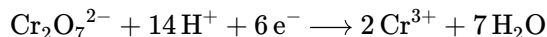
- **Acidic** (high H^+) → equilibrium shifts right → **dichromate**, $\text{Cr}_2\text{O}_7^{2-}$, **orange**.

- **Alkaline** → shifts left → **chromate**, CrO_4^{2-} , **yellow**.

Structures

Chromate CrO_4^{2-} : **tetrahedral**, yellow. **Dichromate** $\text{Cr}_2\text{O}_7^{2-}$: **two tetrahedra sharing one corner** through a Cr-O-Cr bridge, bond angle **126°**, orange.

Oxidising action in acidic medium



Combine this reduction half with any oxidation half, matching electrons:

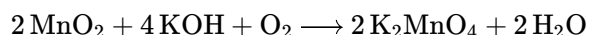
- $6\text{Fe}^{2+} \longrightarrow 6\text{Fe}^{3+} + 6\text{e}^- \rightarrow \text{Cr}_2\text{O}_7^{2-} + 6\text{Fe}^{2+} + 14\text{H}^+ \longrightarrow 2\text{Cr}^{3+} + 6\text{Fe}^{3+} + 7\text{H}_2\text{O}$
- $\text{I}^- \rightarrow \text{I}_2$, $\text{H}_2\text{S} \rightarrow \text{S}$, $\text{Sn}^{2+} \rightarrow \text{Sn}^{4+}$ all work the same way.

Potassium dichromate is used as a **primary standard** in volumetric analysis; sodium dichromate is more soluble and preferred in organic oxidations.

Potassium permanganate — *preparation, why it's purple, and both media*

Preparation — two steps

1. Fusion. MnO_2 fused with KOH in air (or with KNO_3 as oxidant) gives green potassium manganate:

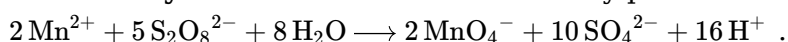


2. Oxidation of manganate to permanganate, either by disproportionation in acid or — commercially — by electrolytic oxidation in alkaline solution:



[exposure] That second reaction is a **disproportionation** — a reaction where *the same element* is simultaneously oxidised and reduced. It's a distinct category because normally one species oxidises while a different one reduces. Here manganese starts at +6 in MnO_4^{2-} and ends up both higher (+7 in MnO_4^-) and lower (+4 in MnO_2) in the same step.

Laboratory route: a Mn^{2+} salt oxidised by peroxodisulphate:



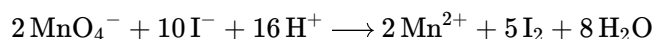
Structure and colour

Both MnO_4^- and MnO_4^{2-} are **tetrahedral**. Manganate (MnO_4^{2-} , Mn is +6, $3d^1$) is **green and paramagnetic** — one unpaired electron. Permanganate (MnO_4^- , Mn is +7, $3d^0$) is **purple and diamagnetic**. Its colour comes from **charge transfer**, not d-d, since it has no d electrons.

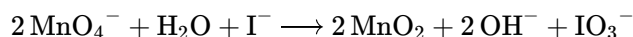
Heating above 513 K decomposes it: $2\text{KMnO}_4 \longrightarrow \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$.

Oxidising action — both media, both asked

Acidic: $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$ (5 electrons)



Alkaline/neutral: $\text{MnO}_4^- \rightarrow \text{MnO}_2$ (3 electrons), and iodide goes to **iodate**, not iodine:



Trap: the products differ by medium. Acidic gives Mn^{2+} and I_2 ; alkaline gives MnO_2 and IO_3^- . Writing the acidic products for an alkaline question loses the whole answer.

Examiner asks: "why is permanganate titration not carried out in presence of HCl?" — KMnO_4 oxidises HCl to Cl_2 , a side reaction giving an inaccurate end point. Use H_2SO_4 instead.

Lanthanoids — *the f block, and the contraction that explains everything*

[exposure] The **lanthanoids** are the 14 elements following lanthanum (Ce, $Z = 58$, to Lu, $Z = 71$) in which the **4f orbital** progressively fills. They're pulled out of group 3 and printed as a separate row because they're so chemically alike that keeping them in the main table would be unwieldy. General configuration: $4f^{1-14} 5d^{0-1} 6s^2$. Their dominant oxidation state is +3, with a few showing +2 or +4 where that gets them closer to an empty, half-filled or full f subshell.

[exposure] **Lanthanoid contraction** is the steady decrease in atomic and ionic radii across the lanthanoid series, from lanthanum to lutetium, as atomic number rises. It happens because 4f electrons **shield poorly** — an f electron is diffuse and doesn't screen the outer electrons from the nucleus effectively. So each added proton pulls the whole electron cloud in a little more, and the atoms shrink steadily instead of staying the same size. The effect is small per element but accumulates over 14 elements into something chemically decisive.

Its three consequences — the reason it's examined so heavily

1. **Lanthanoids are hard to separate.** Similar radii → similar chemical properties → separating a mixture is difficult.
2. **4d and 5d elements have nearly identical sizes** (Zr and Hf, Nb and Ta), because the contraction cancels the expected size increase.
3. **Second and third transition series resemble each other** far more than either resembles the first.

Eu^{2+} **is a strong reducing agent** — +3 is the stable lanthanoid state, so Eu^{2+} readily loses an electron to reach it, reducing something else in the process.

[exposure] **Misch metal** is a commercially useful lanthanoid alloy — roughly **95% lanthanoid, 5% iron**, with traces of S, C, Ca and Al. It exists because separating individual lanthanoids is expensive and often unnecessary; the mixture works fine. It's used in magnesium-based alloys and to make **bullets, shells and lighter flints**, since it sparks when struck.

Actinoids — and the three-way comparison with lanthanoids

[exposure] The **actinoids** are the 14 elements after actinium (Th, $Z = 90$, to Lr, $Z = 103$) in which the **5f orbital** fills. What sets them apart from lanthanoids is that **all of them are radioactive**, and their 5f, 6d and 7s orbitals are so close in energy that electrons from all three join in bonding — giving a much wider range of oxidation states (+3 through +6 and beyond) than the lanthanoids' near-uniform +3. That, plus the radioactivity, is why their chemistry is described as more complicated.

Actinoid contraction is greater than lanthanoid contraction, because 5f electrons shield even more poorly than 4f, so effective nuclear charge rises faster and the radii shrink more sharply.

The three differences to write

	Lanthanoids	Actinoids
Orbital filling	4f	5f
Shielding	Poor	Even poorer → greater contraction
Oxidation states	Mainly +3; some +2, +4	+3 common, but also +4, +5, +6
Radioactivity	Only promethium	All of them

Examiner asks: "three points of difference" (2020) — pick orbital, contraction magnitude, and oxidation-state range. Radioactivity is the easy fourth if you need it.

Numerical and equation patterns, collected — the calculable parts of a mostly-descriptive chapter

A · Spin-only magnetic moment — 2–3 marks

Recognise it: an atomic number is given, or "calculate the magnetic moment".

1. Write the neutral atom's configuration from Z .
2. Remove electrons for the ion — **4s first**, then 3d.
3. Fill the five d orbitals singly before pairing; count unpaired = n .
4. $\mu = \sqrt{n(n+2)}$ BM.

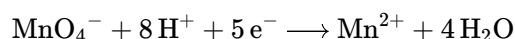
B · Balancing a redox equation — 3–5 marks

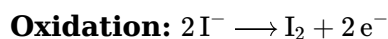
Recognise it: "complete and balance", usually with MnO_4^- or $\text{Cr}_2\text{O}_7^{2-}$ in acidic medium.

1. Write reduction and oxidation halves separately.
2. Balance the metal, then O with H_2O , then H with H^+ .
3. Balance charge with electrons.
4. Scale both halves to equal electrons, add, cancel.

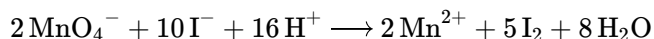
Worked · 2025 — permanganate and iodide in acid

Reduction: $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$. Add 4 H_2O right, 8 H^+ left. Charge: left $-1 + 8 = +7$, right +2. Add 5 electrons left:





Multiply reduction by 2, oxidation by 5:



C · Oxidation number from a formula — 1-2 marks

Recognise it: "write the oxoanion in which the metal shows a state equal to its group number".

1. Set the metal as x ; oxygen is -2 each; sum to the overall charge.
2. For Mn (group 7): MnO_4^- gives $x - 8 = -1$, so $x = +7$. For Cr (group 6): CrO_4^{2-} or $\text{Cr}_2\text{O}_7^{2-}$, both $+6$.

D · Identify the compound (A, B, C chains) — 5 marks

Recognise it: a colour-coded sequence — "black-brown solid A fused with KOH gives dark green B, which on oxidation gives dark purple C".

1. Match colours to compounds: black-brown $\text{MnO}_2 \rightarrow$ dark green $\text{K}_2\text{MnO}_4 \rightarrow$ dark purple KMnO_4 .
2. Write each step's equation.

Past year questions · question types — ranked by how often they turn up

From the last 10 years of boards — 25 questions, grouped by pattern. This chapter is overwhelmingly reason-based: most marks come from explaining, not calculating.

1 · "Give reason" on configuration stability — 2-3 marks

Recognise it: any comparison of two ions, oxidising/reducing strength, or an E° anomaly.

1. Write both configurations, before and after electron transfer.
2. Point at whichever is d^5 or d^{10} .
3. State the consequence in the question's own language.

Trap: answering "because it's stable" without naming *which* configuration. The $3d^5$ is the mark.

2 · Balancing redox equations — 3-5 marks

Recognise it: "complete and balance the following".

1. Halves separately.
2. Metal $\rightarrow$ O with water $\rightarrow$ H with H^+ $\rightarrow$ charge with electrons.
3. Scale, add, cancel.

3 · Lanthanoid / actinoid comparison — 3 marks

Recognise it: "lanthanoid contraction", "three differences", "why is actinoid chemistry more complicated".

1. Name the orbital (4f vs 5f).

2. Poor shielding → contraction; 5f worse than 4f.
3. Add the oxidation-state range and radioactivity.

4 · Magnetic and colour properties — 2-3 marks

Recognise it: "which is colourless/paramagnetic and why", or a μ calculation.

1. Get the ion's d-configuration.
2. Count unpaired electrons.
3. None → colourless and diamagnetic. Some → coloured and paramagnetic.
4. Check for the charge-transfer exception (KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$).

5 · KMnO_4 / $\text{K}_2\text{Cr}_2\text{O}_7$ preparation and reactions — 3-5 marks

Recognise it: named reagent questions, or the A/B/C identification chain.

1. Preparation equations.
2. The pH interconversion for chromate/dichromate.
3. Oxidising action — check which medium is asked.

6 · Physical-property trends — 2-3 marks

Recognise it: enthalpy of atomization, melting point, or a graph across the series.

1. Count unpaired electrons.
2. More unpaired → stronger metallic bond → higher value.
3. For 2nd/3rd series, add the poor-shielding argument.

Past year questions · mark slots — *what each type is worth*

Question type	Slot	Time
"Give reason" (configuration)	2-3 marker	2-3 min
Balancing redox equations	3-marker; both media is a 5	5-7 min
Lanthanoid/actinoid comparison	3-marker	3 min
Magnetic moment calculation	2-3 marker	3 min
$\text{KMnO}_4/\text{K}_2\text{Cr}_2\text{O}_7$ chemistry	3-5 marker	5-7 min
Graph/trend interpretation	3-marker	4 min
Definitions (misch metal, oxoanion)	1-2 marker	2 min

The video splits these as short-answer type I (2 marks), type II (3 marks) and long answer (5 marks) — the KMnO_4 chemistry and multi-part "answer the following" sets are where the 5-markers sit.

Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

2017 Q3 · 2017 Q8 · 2019 Q2

"Transition metals show variable oxidation state" and "Zn, Cd, Hg are soft metals" — asked together, repeatedly, in the same two-part format.

2024 Q5 · lecture PYQ

Define lanthanoid contraction, then why actinoid contraction is greater. The single most reliable f-block question.

2025 Q4 · 2026 Q25(b)

Balance MnO_4^- with I^- — once in acidic medium alone, once acidic *and* alkaline in the same question.

2022 Q15 · 2022 Q16 · lecture PYQ

Graph questions on enthalpy of atomization and E° values — lowest in 3d series (Zn), why 2nd/3rd series are higher, why Mn's E° is exceptionally negative, why Cu's is exceptionally positive.

2020 Q11 · 2026 Q22

Lanthanoids vs actinoids — three differences, then why actinoid chemistry is more complicated.

2020 Q10 · 2022 Q12 · 2020 Q6

Colour and magnetism: Sc^{3+} vs Cr^{3+} , which ion is colourless, spin-only moment of Co^{2+} . All answered by counting unpaired electrons.

2023 Q17 · 2017 Q13 · 2020 Q20

Mn's +7 state, transition metals as catalysts, and interstitial compound formation. Standard reason-based trio.

Past year questions · cold practice — *answers only* — *work them before you look*

Configuration reasoning

- 2026 Q7(a) — why is the third ionization enthalpy of Mn high? Mn^{2+} is $3d^5$, half-filled and stable; removing a third electron breaks it
- 2019 Q13(c) — why is $E^\circ(\text{Mn}^{3+}/\text{Mn}^{2+})$ highly positive compared with $\text{Cr}^{3+}/\text{Cr}^{2+}$? Mn^{3+} gains one to reach stable $3d^5$; Cr^{3+} is already stable $3d^3$
- 2017 Q9(a) — how does variability of oxidation state differ between transition and p-block? Transition differ by 1 unit; p-block by 2
- 2026 Q21(a) — why is Cu a transition element despite a filled d orbital? Cu^{2+} is $3d^9$ — partially filled in its common oxidation state

Magnetism and colour

- 2020 Q6 — spin-only moment of Co^{2+} , $Z = 27$. $3d^7$, $n = 3$, $\mu = 3.87$ BM
- 2022 Q12 — which of Ti^{4+} , Cr^{3+} , V^{3+} is colourless and why? Ti^{4+} — $3d^0$, no d-d transition possible

- 2020 Q10 — Sc^{3+} or Cr^{3+} : which shows magnetic behaviour? Cr^{3+} ($3d^3$, paramagnetic); Sc^{3+} is $3d^0$, diamagnetic
- 2026 Q21(b) — KMnO_4 or K_2MnO_4 : which is paramagnetic? K_2MnO_4 — Mn is +6, $3d^1$, one unpaired electron

Equations

- 2025 Q4 — balance $\text{Cr}_2\text{O}_7^{2-}$ with Fe^{2+} in acid.

$$\text{Cr}_2\text{O}_7^{2-} + 6\text{Fe}^{2+} + 14\text{H}^+ \longrightarrow 2\text{Cr}^{3+} + 6\text{Fe}^{3+} + 7\text{H}_2\text{O}$$
- 2026 Q25(b) — MnO_4^- with I^- in alkaline medium.

$$2\text{MnO}_4^- + \text{H}_2\text{O} + \text{I}^- \longrightarrow 2\text{MnO}_2 + 2\text{OH}^- + \text{IO}_3^-$$
- 2026 Q25(a) — identify A, B, C: black-brown solid fused with KOH in air $\rightarrow$ dark green $\rightarrow$ electrolytic oxidation $\rightarrow$ dark purple. A = MnO_2 , B = K_2MnO_4 , C = KMnO_4
- 2023 — what happens when KMnO_4 is heated? $2\text{KMnO}_4 \longrightarrow \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$ at 513 K

f block

- 2019 Q19 — why is Eu^{2+} a strong reducing agent? Loses one electron to reach the stable +3 lanthanoid state
- 2019 Q19(b) — why is separating a lanthanoid mixture difficult? Similar radii from lanthanoid contraction $\rightarrow$ similar chemical properties
- 2019 Q14(a) — general electronic configuration of lanthanoids. $4f^{1-14} 5d^{0-1} 6s^2$
- 2025 Q24(a) — what is misch metal, and one use? $\sim 95\%$ lanthanoid + 5% Fe with traces; used in Mg alloys, bullets, shells, lighter flints

Built from Sourabh Raina's d and f Block one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 4 (Rationalised 2022-23). First-contact material: every term marked **[exposure]** is explained from scratch before use.